

Vinay Kumar Chopra

3rd Year Computer Science Student · BITS Pilani (Online) via Scaler

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SUMMARY

Motivated 3rd year CS student with hands-on experience building **backend systems** and **data tools** in C++, Java, and Python. Strong foundation in low-level programming, REST API design, and database management. Actively seeking a *software development internship* to apply and deepen these skills in a real-world engineering environment.

EDUCATION

Birla Institute of Technology, Pilani (BITS Pilani)

Expected 2027

Bachelor of Science (Hons.) in Computer Science · Online Program
Coursework via Scaler School of Technology · CGPA: 8.18 / 10 · 3rd Year

SKILLS

Languages	C++, Java, Python, SQL
Frameworks	Spring Boot, Hibernate, JPA, Flask, Django
Tools	Git, Docker, REST APIs, Jira, Pandas, Matplotlib
Concepts	Machine Learning, Low-Level Programming, Schema Design, Concurrency

PROJECTS

Market Data Publisher

C++ · Sockets · Concurrency

- Developed a **high-performance C++ system** for low-latency market data dissemination.
- Implemented concurrent **TCP/UDP socket programming** for efficient real-time data streaming.
- Optimized data structures and serialization methods to **minimize latency**.

Product Service

Java · Spring Boot · Hibernate

- Built a scalable **Spring Boot application** with full CRUD functionality, improving data persistence by **40%** using Hibernate and JPA.
- Designed RESTful APIs for e-commerce and inventory management, achieving a **25% increase** in data retrieval speed.
- Conducted end-to-end testing across all modules, reducing reported post-deployment issues by **30%**.

Aadhar Seva Radar

Python · Flask · Django

- Developed a **Python web application** to streamline search and management of Aadhar-related service information.
- Implemented efficient query mechanisms, advanced data filtering, and designed the **full database schema**.

Blood Report Analyzer

Python · Pandas · Matplotlib

- Built a tool to **analyze and visualize digital blood report data** with automated reference range checking and trend identification.

OPEN SOURCE CONTRIBUTIONS

- Contributed to **OSGeo GRASS GIS** (Google Summer of Code repo) — merged pull requests *PR#7097* and *PR#7005*.
- Contributed to **storacha/guppy** (PLDG project) — merged *PR#195*.